



Emotion Recognition with Facial Landmark Heatmaps

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Abstract. Facial expression recognition is a very challenging problem and has attracted more and more researchers' attention. In this paper, considering that facial expression recognition is closely related to the features of key facial regions, we propose a facial expression recognition network that explicitly utilizes the landmark heatmap information to precisely capture the most discriminative features. In addition to directly adding the information of facial fiducial points in the form of landmark heatmaps, we also propose an end-to-end network structure--heatmap aiding emotion network (HAE-Net) by embedding the landmark detection module based on stack-based hourglass network into the facial expression recognition network. Experiments on CK+, RAF and AffectNet databases show that our method achieves better results compared with the state-of-the-art methods, which demonstrates that adding additional landmark information, as well as joint training of landmark detection and expression recognition, are beneficial to improve recognition performance.

Keywords: Facial expression recognition · Facial landmark detection · Facial landmark heatmaps

1 Introduction

Emotion recognition aims to identify the expression from an input facial image, and the classic problem is to classify it as one of the seven basic expressions--angry, disgust, fear, happy, sad, surprise and neutral [12]. Recently, it has attracted more and more attention from both academic and industrial communities due to its wide range of applications in many fields such as human-computer interaction and facial animations.

With the development of deep learning algorithms, a series of networks such as VGG [23], Resnet et al. [3], have been proposed and widely used in image classification [7], face recognition [21] and many other tasks. Since the convolutional neural network (CNN) can learn the high-level feature representation of

the image, many researchers built up facial expression recognition systems based on it and achieve great performance. For example, Ali et al. [15] proposed a deep neural network architecture based on Inception layer [20] for facial expression recognition problem. These methods mainly extract spatial information, but do not consider the characteristics associated with expressions.

As a special classification task related to facial analysis, facial expression specific prior knowledge in facial images can be used to boost the recognition performance. For example, according to the fact that facial expression is mainly caused by changes of fiducial points location, [5] designed a temporal geometry network that used normalized coordinates of facial landmarks from consecutive frames to learn temporal geometric features. However, using sequences of facial landmarks requires to collect more data and it is not suitable for static image recognition.

In this paper, we mainly focus on the task of static facial expression recognition. Since the expression process of the expression is closely associated with the movements of the facial key parts such as eye, nose and mouth, et al., it is possible to improve the recognition accuracy by capturing the features of these parts through the facial fiducial point positions. Based on this, we design an expression recognition system that can simultaneously utilize appearance information and geometry information. Figure 1 illustrates the proposed framework. We fed the facial landmark heatmaps to the network as extra aiding information while using the CNN network to extract texture features from facial images.

Different from using sequences of facial landmark points [5, 24, 28, 30], we only pay attention to facial fiducial points of a single image expressing peak emotion. Besides, instead of directly using the coordinates of facial landmark points like [13], we add the geometry information in the form of landmark heatmaps to the appropriate layer of the network. Here the heatmaps are generated by the Gaussian kernels centered on the locations of facial landmarks, indicating the probability of occurrence of each component. Thus adding a heatmap map is equivalent to increasing facial structure information, including the location information of the facial organs, visibility and the relationship between them.

In summary, the main contributions of this paper are:

- (1) We propose to exploits facial landmark heatmaps as geometry prior knowledge in a convolutional neural network for the facial expression recognition task. Through these heatmaps, we can focus on the most discriminative part of the extracted features;
- (2) We design an end-to-end network that uses predicted heatmaps estimated by a facial landmark heatmap detection network to refine the expression features, and we call it Heatmap Aiding Emotion Network;
- (3) Experiments on three databases achieve better accuracy than other state-of-art methods.

2 Related Work

The method we propose is mainly related to the knowledge of the expression recognition using geometry information and facial landmark detection network. We first introduce the relevant work and then illustrate the motivation of our work.

2.1 Expression Recognition Using Facial Landmarks

To improve the accuracy of expression recognition, most approaches tend to add prior knowledge. Among them, facial landmark points prior knowledge is added most as the facial key parts surrounded by them contain the most descriptive information related to expression.

Many methods use facial landmark prior knowledge to get better recognition results. One common practice is to use coordinates of fiducial points as input and apply a neural network to extract features, which is called geometry features. For example, Zhang et al. [30] proposed a part-based hierarchical bidirectional recurrent neural network (PHRNN) to simulate facial morphological variation and dynamic changes of expression by gradually adding landmark information of a single face organ in each BRNN layer. Some also design relevant new feature representations using coordinates of facial landmark points. Tang et al. [24] design a 39-dimensional feature based on the coordinates of facial landmark points to represent the facial geometry information of a single frame. Another approach is to develop multi-task learning methods to simultaneously perform expression recognition and facial landmark detection. For example, Hu et al. [4] proposed an end-to-end soft sharing strategy that automatically learned which layers to share and improved the recognition accuracy by using an auxiliary task: facial landmark detection; Tautkute et al. [25] expanded the latest facial landmark detection network—Deep alignment network [6] to a multi-task network by adding a classification loss function. Then the facial image can be classified into the corresponding expression category while performing facial landmark detection.

In recent years, the landmark information has been added to the network in the form of a heatmap, which has made great progress in face super-resolution reconstruction [1]. Inspired by these work, we believe that if landmark heatmap information is added in the process of extracting image texture features, it can provide auxiliary information to enhance the expression recognition performance. Different from the above methods that directly use coordinates of facial landmark points, we first exploit facial landmark points to generate the ground-truth heatmaps and then concatenate the convolutional feature maps extracted from the original image with the landmark heatmaps to emphasize the features at the locations of facial key parts.

2.2 Landmark Heatmap Regression

Heatmap regression models, which aim to predict landmark heatmaps, have achieved excellent performance in tasks such as pose estimation [17] and face alignment [27]. Motivated by this, to enable end-to-end training, we incorporate the heatmap estimate network module to our network, as shown in Fig. 3.

Since one of our goals is to generate the likelihood heatmaps for facial fiducial points, we employ the network in [17], which consists of stacked hourglass modules. As shown in [17], the stacked-hourglass network achieved a compelling

result as it is built by a repeated bottom-top and top-bottom mechanism which enables the network to fuse multi-scale features.

One of our contributions is to show that when integrating a facial landmark heatmap estimation network to an expression-recognition network and training them jointly, we can get higher recognition accuracy.

3 Proposed Method

We first introduce the framework of the proposed method that exploits extra facial landmark prior information, and then describe the end-to-end network structure, which we call Heatmap Aiding Emotion Network (HAE-Net).

3.1 Framework

The overall framework of our proposed method is shown in Fig. 1. We first use the feature extraction part to get the feature maps from the original images. Then we generate the facial landmark heatmaps with facial detection module. Finally, both convolutional feature maps and landmark heatmaps are concatenated and fed into the identification net to get the final predicted expressions.

To better illustrate the effectiveness of the jointly training of facial expression and landmark heatmap prediction, we also first design a basic network—Conv_heatmap that uses the ground-truth heatmaps directly. It contains two inputs, one is the facial image and the other is the ground-truth heatmap. Comparative experiments can be seen in the fourth part of the paper.

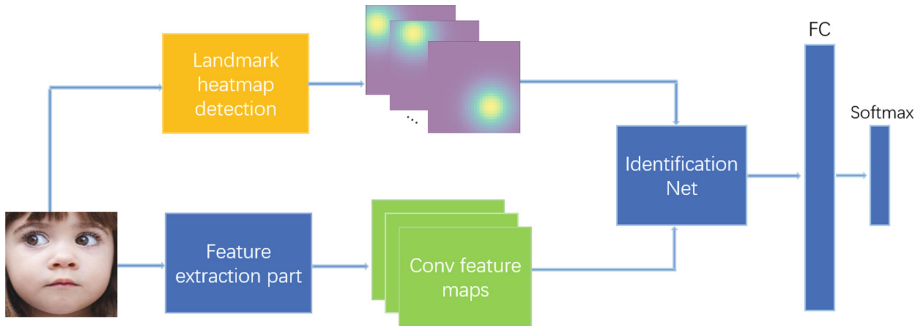


Fig. 1. The framework of our method.

We choose to leverage the heatmap information for two reasons. First, using a convolutional neural network only extracts the texture features from original images, but using extra landmark heatmaps as prior knowledge adds facial shape information [2]. Second, the landmark heatmap not only provides the locations of facial fiducial points but also indicates the visibility of each point around

facial components and the relationships between them [29], which is beneficial to extract the features of the facial parts closely related to the expression. As shown in Fig. 2, the landmark heatmap of 37 channels and the original image are shown together in a single image.

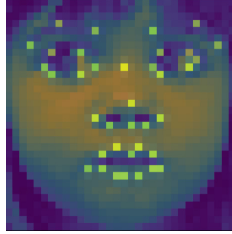


Fig. 2. The generated heatmap using 37 facial landmarks. We visualize it in a single image by adding up the 37 channels of the generated heatmaps.

3.2 Heatmap Aiding Emotion Network

Figure 3 illustrates the structure of our proposed Heatmap Aiding Emotion Network (HAE-Net). HAE-Net consists of two parts: a landmark heatmap prediction branch and a facial expression recognition branch. Each orange block represents one or more residual blocks in the top landmark heatmap prediction branch. The blue block represents the convolutional layer, while the yellow block indicates the fully connected layer. Particularly, the batch normalization layer, ReLu layer and max-pooling layer are not drawn here. The entire network can be trained in an end-to-end manner.

We build up the facial expression recognition branch based on [8], which consists of six convolution layers followed by batch normalization layer, ReLu layer, max-pooling layer and two fully connected layers. We name the first convolution layer as conv_1, and so on. As the features of the lower layers facilitate landmarks location [18], we decide to add landmark information at the lower layers of the network. We use conv_1 to conv_3 as the feature extraction part, which will get a feature map of size 32×32 . Then instead of directly predicting locations, we adopt the heatmap detection network built by stacked hourglass networks to predict the probability heatmaps of each facial landmark. The heatmap estimation network is mainly composed of four stacked hourglass modules, each of which outputs a predicted landmark heatmap for supervising the training of the network. Specifically, the hourglass network is based on an encoding-decoding structure, in which the features are first downsampled to a low resolution using convolutional and max-pooling layers and then upsampled to the original resolution. Meanwhile, features extracted in the downsampling layers are combined with the features at the upsampling layers. In this way, the network can capture effective information from features of multiple scales. At last, the heatmaps obtained through the landmark detection network are combined with the lower

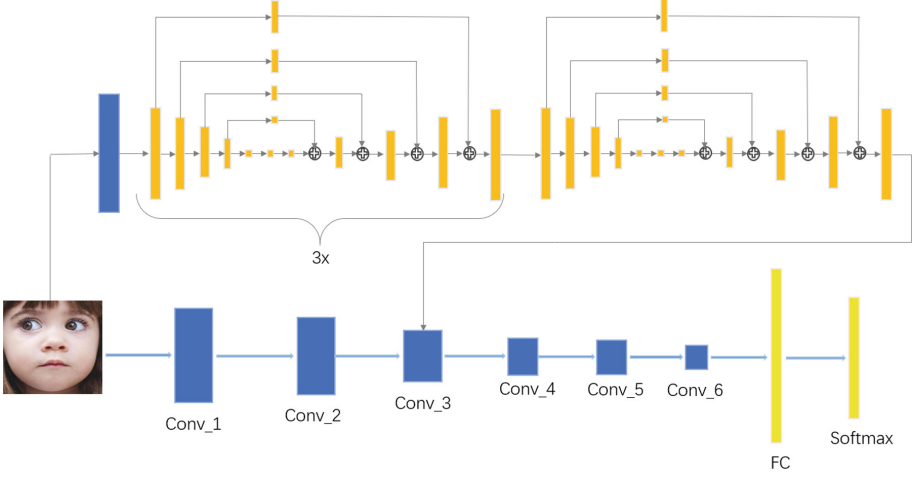


Fig. 3. The structure of Heatmap Aiding Emotion Network. HAE-Net consists of two parts: a landmark heatmap prediction branch (top) and a facial expression recognition branch (bottom).

layer features extracted by the feature extraction part to further extract the high-level features to improve the recognition performance.

By adding intermediate supervision in the stacked hourglass networks, the heatmap loss is as follows:

$$L_{Heatmap_loss} = \sum_{i=1}^n \|H_i - H^*\|_2 \quad (1)$$

where H_i is the i -th output of hourglass modules, and H^* is the ground-truth generated heatmap.

Cross-entropy is used to supervise the learning of expression classification, namely:

$$L_{Softmax_loss} = -\frac{1}{N} \left(\sum_{i=1}^N y_i \cdot \log(f(x_i)) \right) \quad (2)$$

where N is the total number of samples, x_i represents the i -th sample, y_i is the true label, and $f(x_i)$ is the predicted output. Thus, the final loss function of the network is:

$$L_{Total_loss} = L_{Softmax_loss} + \lambda \cdot L_{Heatmap_loss} \quad (3)$$

where λ is used to balance the influences of heatmap loss and softmax loss.

With this method, we can not only locate the landmarks but also exploit the landmark information to improve facial expression recognition performance.

4 Experiments

In this section, datasets and implementation details are first introduced. Then comparative experiments are conducted on three different datasets to demonstrate the effectiveness of our proposed method.

4.1 Datasets

CK+ [12]: This dataset contains 593 image sequences of 327 subjects. Among them, only 327 sequences with 118 subjects are labeled as one of the seven expressions. We remove the sequences labeled as “contempt” and only use the first (neutral expression) and the last three frames of the rest sequences, which results in 1308 images. We also select 37 landmarks accordingly from original landmark annotations for our experiments. The images are split into 10 folds without subject overlap based on the subject identity in an ascending order. We use the average of the 10 runs as the final accuracy.

RAF [8]: This dataset contains around 30k facial images under natural conditions downloaded from the Internet. It provides two types of images: basic expressions and composite expressions, as well as their corresponding 5 landmark locations labeled manually and 37 landmark locations automatically annotated with the face++ API [14]. In this paper, only images labeled with seven basic expressions categories and 37 landmark annotations are used, which contain 12271 training samples and 3068 testing samples.

AffectNet [16]: This dataset is the largest database for emotion recognition that contains more than 1,000,000 images, about half of which are manually annotated for the presence of seven discrete facial expressions, 68 facial landmark locations and the intensity of valence and arousal. Here we only select images with seven basic expressions and pick out 37 landmarks from original landmark annotations for our experiments.

4.2 Implement Details

For CK+ and RAF dataset, we crop the face region based on the ratio of inter-pupil distance. For Affect dataset, we crop the face region according to the bounding box coordinates. Then the cropped faces are aligned according to the center coordinates of the left and right eyes and resized to the same size $128 * 128$. We also use 37 landmarks to generate the heatmaps as extra input or ground truths for training. We implement data augmentation by random cropping, rotating and flipping. The model is trained using RMSprop optimizer with an initial learning rate of $2.5e-4$. To train the HAE-Net, we set the coefficient λ empirically as 50 for all of the datasets.

4.3 Results and Analysis

We measured the performance of the proposed network architecture on three datasets: CK+, RAF and AffectNet. To measure the results more comprehensively, we choose two metrics: accuracy which is defined as the proportion of the

correct results to all predictions and average accuracy which can be obtained by calculating the average diagonal value of the confusion matrix.

Result of CK+ Dataset. As shown in the Table 1, the accuracy of Conv_Heatmap is about 1% higher than Baseline, which indices that adding the landmark heatmaps helps refine the features. HEA-Net gets about 1% higher accuracy compared with Conv_Heatmap. We owe this to the jointly training of facial landmark detection and facial expression recognition.

The results of our comparison with other methods are listed in Table 1. HEA-Net surpasses most methods by a large margin. However, we do not outperform the method Lopes et al. [11] and Hu et al. [4]. The explanations are as follows. Lopes et al. [11] use extra synthetic samples to train the network, whereas we only train our network on the original database. Although the accuracy of our method is lower than Lopes et al. [11], the average accuracy of HAE-Net is comparable to it, which demonstrates that our proposed method is useful for all seven expressions. Hu et al. [4] pretrained the network on the large dataset-LSEMSW and then used the augmented CK+ database to finetune the network, while we train our network all from scratch.

Table 1. Accuracy and average accuracy on CK+ dataset.

Methods	Validation settings	Accuracy	Average accuracy
Rivera et al. [22]	8-fold	0.8930	0.8594
LBP+SVM [19]	10-fold	0.9140	0.8914
AUDN [10]	10-fold	0.9370	-
Lopes et al. [11]	8-fold	0.9880	0.9575
Hu et al. [4]	5-fold	0.9640	-
Baseline	10-fold	0.9436	0.9427
Conv_Heatmap	10-fold	0.9583	0.9484
HEA-Net	10-fold	0.9624	0.9557

The confusion matrix of HAE-Net on the CK+ Dataset is displayed in Fig. 4. The expression angry gets relatively low accuracy compared with other emotions and it is easily classified into the neutral expression. The reason may be that the angry expression is closely related to the action unit frowning, but there is no facial landmarks representing the position between the eyebrows. The features most related to the angry may not be captured well, thus it will be easily classified into the neutral expression with most training samples.

Result of RAF Dataset. The results of the different methods on the RAF database are shown in Table 2. We train all of our networks from scratch using training samples and evaluate the performance on testing samples. Comparing

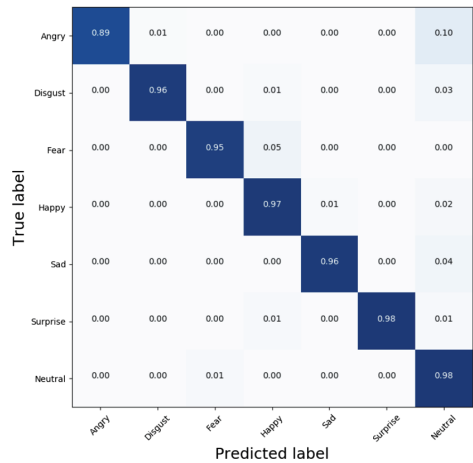


Fig. 4. Confusion matrix of HAE-Net on the CK+ dataset.

the results of the Baseline with Conv_Heatmap, it can be seen that adding the landmark heatmap can bring about performance improvement. Further, the result of HAE-Net surpasses that of Conv_Heatmap, which shows that jointly training landmark heatmap detection and expression recognition is beneficial to improve the accuracy of expression recognition. FSN [31] proposed a feature selection mechanism to filter unrelated features, while gACNN [9] learned adaptive weights for different key facial patches extracted by facial landmarks and the whole image and then integrates the local representations with global representations. As is shown in Table 2, the results of our method are better than any of them.

Table 2. Accuracy and average accuracy on RAF dataset.

Methods	Accuracy	Average accuracy
FSN [31]	-	0.7246
gACNN [9]	0.8507	-
Baseline	0.8478	0.7377
Conv_Heatmap	0.8550	0.7564
HEA-Net	0.8664	0.7671

Table 3. Accuracy and average accuracy on AffectNet dataset.

Methods	Accuracy	Average accuracy
RAN-ResNet18 [26]	0.5297	-
gACNN [9]	0.5878	-
Baseline	0.5620	0.5628
Conv_Heatmap	0.5769	0.5757
HEA-Net	0.5886	0.5871

From the confusion matrix shown in Fig. 5, we can observe that the accuracy for happy is the highest, but the accuracy of fear is the lowest. We can also find that disgust expression is difficult to distinguish from angry and the fear expression is confused most with sad and neutral.

Result of AffectNet Dataset. The results of AffectNet dataset are presented in Table 3. Similar to the case of RAF dataset, our proposed HAE-Net achieves the best accuracy of all the methods, which verifies the superiority of our method.

The confusion matrix of HAE-Net is shown in Fig. 6. We note that in the case of happy and neutral, our network performs well, but the performance of disgust and surprise is relatively poor. Particularly, neutral and surprise, angry and neutral, disgust and angry, sad and neutral are easily confused with each other in this dataset.

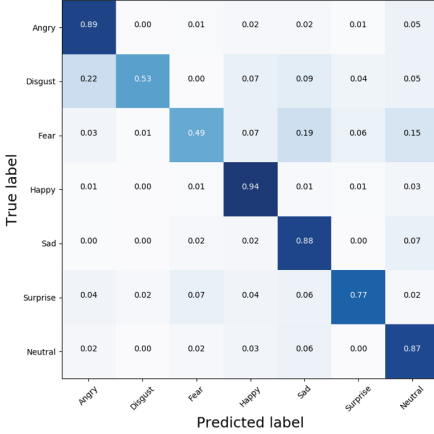


Fig. 5. Confusion matrix of HAE-Net on the RAF dataset.

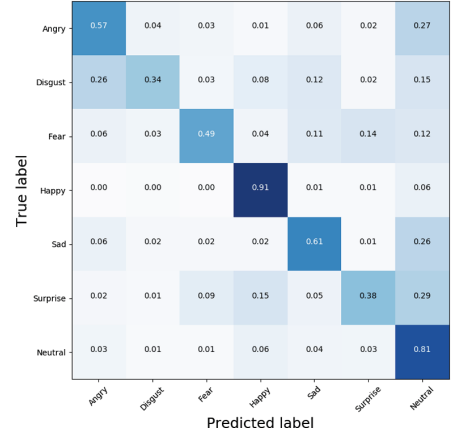


Fig. 6. Confusion matrix of HAE-Net on the AffectNet dataset.

5 Conclusion

In this work, we propose an expression recognition method that exploits landmark heatmap information explicitly and further present an end-to-end expression recognition network structure -- HAE-Net which is embedded with a facial landmark detection module. We conduct comparative experiments on three databases and get compelling results, which show that (1) directly adding additional landmark information helps to capture the discriminate features related to emotions; (2) HAE-Net can further improve the recognition accuracy by jointly training the facial landmark detection and expression recognition.

Acknowledgement. This work was partially supported by the National Natural Science Foundation of China (No. 61471216 and No. 61771276) and the Special Foundation for the Development of Strategic Emerging Industries of Shenzhen (No. JCYJ20170307153940960 and No. JCYJ20170817161845824).

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